

Total No. of Questions : 4]

SEAT No. :

PF-321

[Total No. of Pages : 1

**Apr-26/TE/Insem-406**  
**T.E. (E & TC)**  
**CELLULAR NETWORK**  
**(2019 Pattern) (Semester - VI) (304192)**

**Time : 1 Hour]**

**[Max. Marks : 30**

**Instructions to the Candidates :**

- 1) Answer the Q1. Or Q2 and Q3 or Q4.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) What is diversity? Explain its types in brief. [5]  
b) Employing the Hata model, compute the median loss at a distance  $d=8$  km, when the carrier frequency  $f_c = 2.1$  GHz,  $h_t=40$  m,  $h_r = 2$  m for a large city. [4]  
c) Illustrate the concept fading and explain Multipath Fading. [6]
- Q2)** a) Explain ground- reflection scenario. [5]  
b) Explain free space propagation model & derive the formula  
$$P_t = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2} .$$
 [5]  
c) Explain in details concept of channel estimation in wireless communication. [5]
- Q3)** a) Explain with neat diagram working of OFDM transmitter. [5]  
b) Write advantages, disadvantages & applications of MIMO-OFDM. [6]  
c) Explain Cyclic Prefix in OFDM. [4]
- Q4)** a) Explain multicarrier transmission. [5]  
b) Define smart antenna and explain the types of smart antenna in brief. [5]  
c) Compare OFDM and MIMO OFDM. [5]

